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StudyPilot

AI-Powered Study Dashboard

CMPE 280 — Web UI Design & Development
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TEAM 14

Team Members

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AI, Backend, Voice, Chat

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Dashboard, UI Components

Disha Jadav
Assignments, Calendar

Kumar Harsh
Todos, Tests, Animations

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PostgreSQL, API, Docker

Problem Statement

Why does StudyPilot need to exist?

> *"College students juggle Canvas, Google Calendar, random note apps, and scattered tools — with no single place to see everything, plan smarter, and get help instantly."*

Fragmented Tools

Canvas for assignments, Google Cal for events, note apps for todos — never in sync

Mental Overload

Deadlines tracked in their heads, leading to missed work and poor time management

No Accessibility

Most tools require keyboard & mouse — unusable for students with motor disabilities

No Intelligent Help

Static apps can't analyze your workload and suggest what to study next

Our Solution

StudyPilot is a unified AI-powered study companion that replaces the scattered mess of student productivity tools with one intelligent dashboard.

Student → **Dashboard** → **AI Engine** → **Response**

Todo Manager

Full CRUD + AI task suggestions

Assignment Tracker

Status tracking + AI tutor per assignment

Calendar

Month grid + agenda view + events

AI Chat

Streaming GPT-5.2 with history

Voice Assistant

Hands-free full app control

Dashboard

Unified overview + study plan

Key Features

What makes StudyPilot stand out

01 4 AI-Powered Features

Chat, Study Plan Generator, Assignment AI Tutor, and Todo Suggestions — all backed by GPT-5.2 with real-time SSE streaming so responses appear word-by-word.

GPT-5.2 | Streaming | Azure OpenAI

02 Voice-First Interface

Full hands-free control via "Hey StudyPilot" wake word. Create todos, navigate pages, and ask AI questions — all by voice. Built with Azure Neural STT/TTS.

Azure Speech | Wake Word | Accessibility

03 Secure Full-Stack Auth

JWT authentication with httpOnly cookies, bcrypt password hashing, and complete user-scoped data isolation. Every API route verifies resource ownership.

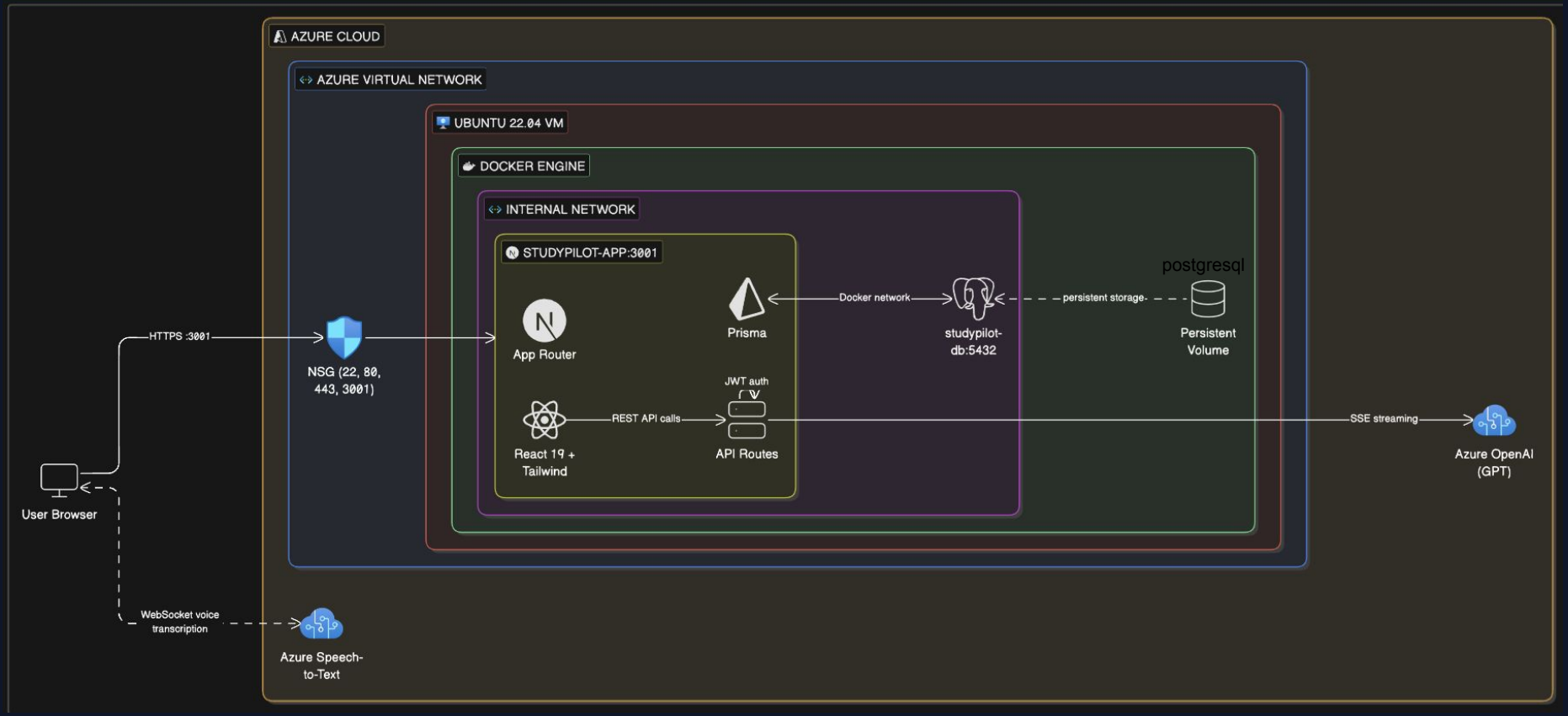
JWT | bcryptjs | Prisma ORM

04 87%+ Test Coverage

315 unit tests across 52 test files using Vitest and React Testing Library. All hooks, components, pages, and context providers are covered.

315 Tests | Vitest | RTL

System Architecture



Tech Stack

Modern, production-grade tools chosen for speed, type safety, and developer experience

FRONTEND

- Next.js 16 (App Router)
- React 19 + TypeScript
- Tailwind CSS v4
- Framer Motion 12
- Radix UI Primitives

BACKEND & DATABASE

- PostgreSQL 16
- Prisma ORM v6
- 19 RESTful API Routes
- JWT + bcryptjs Auth
- Docker Compose

AI & VOICE

- Azure OpenAI GPT-5.2
- Server-Sent Events
- Azure Speech SDK
- Neural TTS JennyNeural
- Wake Word Detection

TESTING & DEVOPS

- Vitest + React Testing Lib
- 315 Tests / 87% Coverage
- Azure Ubuntu VM
- Docker Multi-Stage Build
- GitHub CI

Future Improvements

Where StudyPilot goes next

v1.1 📱 **Native Mobile App**

React Native port with offline support and push notifications for assignment deadlines and calendar reminders

v1.1 🔗 **Canvas LMS Integration**

OAuth2 connection to Canvas to auto-import assignments, grades, and deadlines — no manual entry needed

v1.2 👥 **Collaborative Study Rooms**

Real-time shared workspaces where study groups can co-edit notes, share AI responses, and track group tasks via WebSockets

v1.2 📈 **Study Analytics Dashboard**

Visualize study patterns, completion rates, and focus time with interactive charts. AI coaching suggests habit improvements

v2.0 🧠 **Personalized AI Tutor**

Fine-tuned model trained on each student's notes, syllabus content, and past interactions for truly personalized explanations

v2.0 ♿ **Enhanced Accessibility**

Eye-tracking control, switch access support, dyslexia-friendly fonts, and AI-generated image descriptions for visual content

Demo

StudyPilot

One dashboard. Every tool. Zero switching.